Hospital Bed /OR Scheduling

Smart Hospital Resource Management and Scheduling System

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Team Number: 1 Section: 10

(DEEMED TO BE UNIVERSITY)



Introduction

Efficient hospital resource management ensures limited assets—beds, ORs, staff—are used safely and effectively. Our digital scheduling system coordinates:



Patient admissions



Bed allocation



Operating room availability



Staff schedules priorities

Optimize Resource Allocation

Reduce Patient Waiting Time

Minimize Scheduling Conflicts

Improve Bed P OR Utilization

Support Faster Decision-Making

Flow: Patient → Bed Allocation → OR Scheduling → Staff Coordination → Better Patient Care

Abstract (Key Points)

Purpose

Automate bed & OR scheduling to improve efficiency and reduce conflicts.

Approach

Consider admissions, priorities, availability, capacity, and staff schedules.

Capabilities

Real-time monitoring, conflict minimization, and informed admin decisions.

Benefits

Reduced waiting, better utilization, lower admin workload, scalable design.



Core Inputs: Patient Admissions |Surgery Priority |Bed Availability |OR Capacity |Staff Availability

Outcomes: Less Waiting Time → Better Resource Utilization → Improved Patient Care

Software Requirements

Frontend

HTML, CSS, JavaScript, React.js (if used)

Backend

Node.js, Express.js (if used)

Database

MySQL /MongoDB — To be finalized

Development Tools

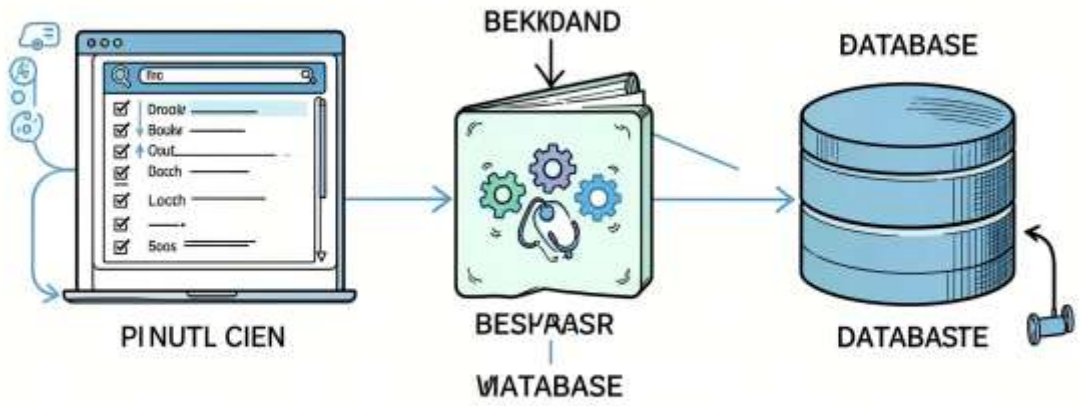
Visual Studio Code, Git / GitHub, Web Browser

Functional Requirements

- User Authentication
- Patient Management
- Bed Management
- OR Scheduling
- Staff Schedule Management
- Real-Time Availability Monitoring
- Admin Dashboard
- Database Management

Architecture

User Interface → Backend/Application → Database



Hardware Requirements

Processor

Intel Core i3 / equivalent or higher

RAM

Minimum 4 GB — Recommended 8 GB+

Storage

Minimum 256 GB

Display

1366 × 768 or higher

Network

Stable LAN / Wi-Fi

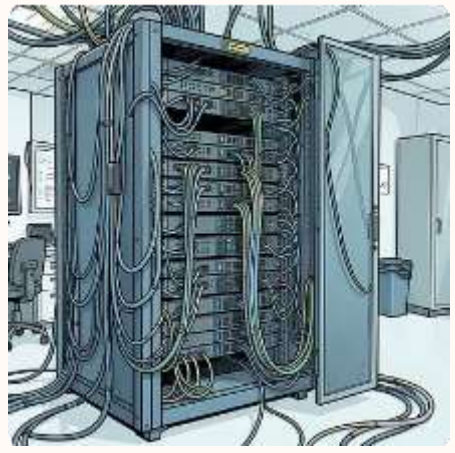
Server

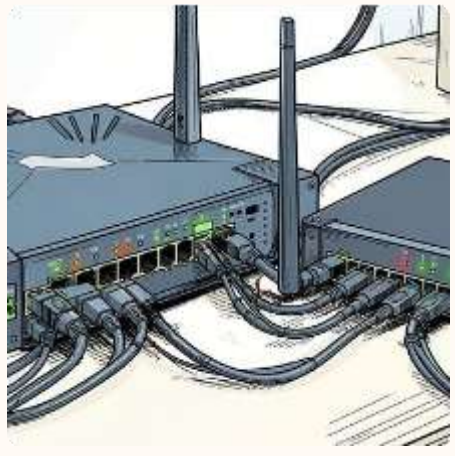
Local or cloud-based server

Optional Hospital Infrastructure

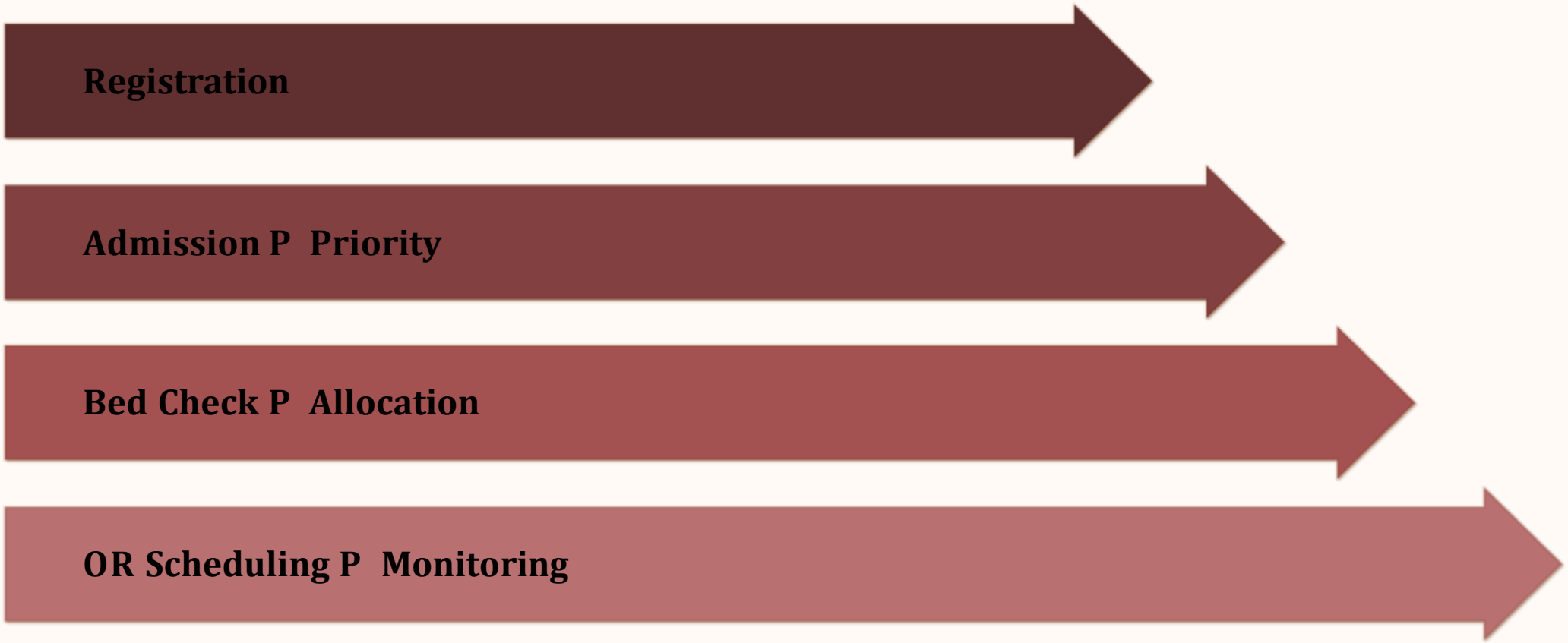
- Administrator workstation
- Nursing station computer
- Dedicated server
- Network router / switch
- Backup storage







Methodology — Workflow P Decision Logic



Major decision factors alongside workflow: Patient Priority · Bed Availability · OR Availability · Surgery Duration · Staff Availability · Emergency Cases

1

Emergency Branch

Emergency cases preempt schedule instant priority reassignment and rapid OR allocation.

2

Real-Time Updates

Live occupancy and schedule updates push notifications to staff and admins.

3

Outcome

Optimized resources · Reduced waiting time · Improved patient care

Literature Review

Research Area	Key Focus	Limitation	Proposed System Advantage
Hospital Bed Allocation Systems	Optimizing bed utilization	Primarily focuses on beds only	Integrates bed allocation with OR scheduling
Operating Room Scheduling	Schedule surgeries by priority & OR availability	Limited integration with other hospital resources	Coordinates ORs, beds, patients, and staff
Healthcare Resource Optimization	Efficient use of limited hospital resources	May require complex optimization algorithms	Provides practical integrated scheduling solution
Real-Time Hospital Management Systems	Monitoring hospital resources	Monitoring alone doesn't optimize schedules	Combines real-time monitoring with scheduling & decision support

Research Gap / Proposed Solution: “The proposed system integrates bed management, OR scheduling, patient priorities, staff availability, and real-time monitoring into a unified hospital resource management platform.”

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